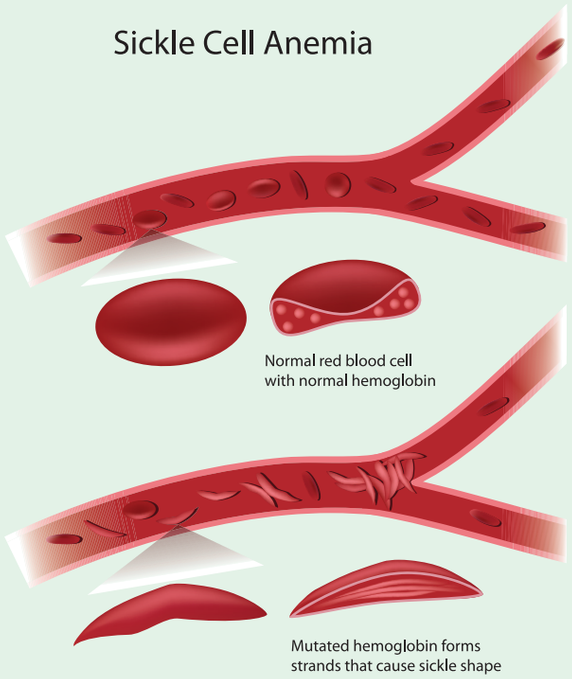
What causes Sickle Cell Anemia?

It is an inherited disease where the body does not produce healthy red blood cells. This is a genetic defect due to a mutation in the β-globin gene of hemoglobin which results in the substitution of a valine in the sixth position, instead of a glutamic acid. This mutation leads to abnormal tertiary structure of haemoglobin which in turn leads to lead to sickle shaped red blood cells. The altered shape of the red blood cells make them rigid and get stuck in the blood vessels and capillaries and thereby slows the blood flow. It is an autosomal recessive disease, meaning that the defective gene must be passed by both the father and mother to the off spring. There is no cure for this disease, but treatments can be given to reduce the symptoms like pain, swelling of limbs, fever etc..



Ramachandran Plots



G.N Ramachandran was born in the town of Ernakulam, Kerala, India to a Tamil speaking family. He completed his B.Sc. honours in Physics from St. Joseph's College, Tiruchirapalli in 1939. He joined the Indian Institute of Science, Bangalore in 1942 in the Electrical Engineering Department. He did his Ph.D. in Physics under the supervision of Nobel Laureate Sir C.V. Raman. In 1942 he received Master's Degree in Physics from Madras University. In 1954, he identified and published the Triple helical structure of Collagen using X-ray diffraction. He pioneered the field of protein structure validation through the study of available crystal structures of peptides. From his studies, in 1962, he developed the Ramachandran Plot which is used even today for stereochemical validation of protein structures. In 1970, he founded the Molecular Biophysics Unit at the Indian Institute of Science which was later known as the Centre of Advanced Study in Biophysics. He has provided a major contribution to the research field of Structural Biology from India for which we must feel proud.

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